E-Detectors

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1 Context and Motivation

In online change detection, two main aspects must be controlled: the occurrence of false alarms and the speed of detection after a change has occurred.

In Algorithm 4, developed previously during the internship, false alarms were controlled using the family-wise error rate criterion. More precisely, under the null hypothesis of no change, the probability that at least one ewe triggers a false alarm was controlled at level α:

$$\mathbb{P}_\infty\left(\text{at least one false alarm occurs}\right) \leq \alpha, \tag{1}$$

where $\mathbb{P}_\infty$ denotes the probability distribution under the hypothesis that no change occurs.

In the present approach, we use another false-alarm criterion called the *average run length* (ARL).

Definition 1. Let N^* denote the stopping time at which the detection procedure raises an alarm. Under the no-change hypothesis, the average run length is defined as

$$\mathrm{ARL}(N^*) := \mathbb{E}_\infty\left[N^*\right]. \tag{2}$$

A detection procedure is said to control the ARL at level α if $\mathbb{E}_\infty\left[N^*\right] \geq \frac{1}{\alpha}$.

Equivalently, its false alarm rate satisfies

$$\mathrm{FAR}(N^*) := \frac{1}{\mathrm{ARL}(N^*)} \leq \alpha.$$

The ARL criterion differs from the family-wise error rate criterion. The latter controls the probability of ever raising a false alarm, whereas the ARL controls the expected time until a false alarm occurs. Therefore, these two criteria should not be interpreted as equivalent.

Among the detection procedures satisfying the prescribed ARL constraint, their performance is evaluated by examining how quickly they detect a change. Two widely used criteria are the worst average detection delay conditioned on the least favourable observations before the changepoint, and the average detection delay conditioned on the event that the procedure stops after the changepoint.

Definition 2 (Worst average detection delays)**.** Let ν denote the unknown changepoint, let $\{\mathcal{F}_n\}_{n\geq 0}$ be the filtration generated by the observations, and define $[x]^+ := \max\{x, 0\}$. The Lorden detection delay is defined by

$$J_{\mathrm{L}}(N^*) := \sup_{\nu\geq 0} \operatorname{ess\,sup} \mathbb{E}_\nu\left[\left[N^* - \nu\right]^+ \,\middle|\, \mathcal{F}_\nu\right]. \tag{3}$$

The Pollak detection delay is defined by

$$J_{\mathrm{P}}(N^*) := \sup_{\nu\geq 0} \mathbb{E}_\nu\left[N^* - \nu \,\middle|\, N^* > \nu\right]. \tag{4}$$

These criteria satisfy $J_{\mathrm{P}}(N^*) \leq J_{\mathrm{L}}(N^*)$.

The general objective is therefore to construct a stopping time N^* with a small detection delay while satisfying the ARL constraint $\mathbb{E}_\infty[N^*] \geq \frac{1}{\alpha}$.

When both the pre-change and post-change distributions are completely known, the classical CUSUM procedure achieves exact minimax optimality with respect to Lorden's criterion among procedures satisfying the same ARL constraint. It is also asymptotically optimal with respect to Pollak's criterion as $\alpha \to 0$. The Shiryaev–Roberts procedure is not exactly minimax optimal withbrespect to Lorden's criterion, but it satisfies strong asymptotic optimality properties.

Motivated by the CUSUM and Shiryaev–Roberts constructions, we consider a more general setting in which the observations are generated by a Poisson process, the pre-change and post-change intensities are not both known, and the observations are not necessarily independent and identically distributed. In this setting, the classical likelihood-ratio CUSUM and Shiryaev–Roberts procedures cannot be applied directly.

Before presenting the proposed algorithms, we first introduce these notions.

2 E-detecteur statistic

2.1 Definitions

Let $\mathcal{F}_n := \sigma(X_1, \ldots, X_n)$ denote the information available up to time n, and let $\mathcal{T}$ denote the set of stopping times with respect to the filtration $\{\mathcal{F}_n\}_{n\geq 0}$.

Let $\mathcal{P}$ be the class of possible pre-change distributions. The notation $\mathbb{P}_{P,\infty}$ and $\mathbb{E}_{P,\infty}$ refers to the case where the observations follow a distribution $P \in \mathcal{P}$ and no change occurs.

Definition 3. [E-process] A nonnegative adapted process $E := \{E_n\}_{n\geq 0}, \qquad E_0 = 1$, is called an e-process for the class $\mathcal{P}$ if

$$\sup_{P\in\mathcal{P}} \mathbb{E}_{P,\infty}[E_\tau] \leq 1, \qquad \forall \tau \in \mathcal{T}. \tag{5}$$

The value E_n measures the evidence against the pre-change model. A large value of E_n indicates that the observations are difficult to explain under the null hypothesis.

Lemma 1. *Let $E = \{E_n\}_{n\geq 0}$ be an e-process for the class of null distributions $\mathcal{P}$. For any $\alpha \in (0,1)$, define the stopping time $\tau_\alpha := \inf\left\{n \geq 1 : E_n \geq \frac{1}{\alpha}\right\}$, with the convention $\inf \varnothing = \infty$. Then*

$$\sup_{P\in\mathcal{P}} \mathbb{P}_P(\tau_\alpha < \infty) \leq \alpha.$$

Equivalently,

$$\sup_{P\in\mathcal{P}} \mathbb{P}_P\left(\sup_{n\geq 1} E_n \geq \frac{1}{\alpha}\right) \leq \alpha.$$

Therefore, rejecting the null hypothesis when E_n first exceeds $1/\alpha$ controls the probability of a false rejection at level α, as in (1).

Proof. Since E is an e-process, $\sup_{P\in\mathcal{P}} \mathbb{E}_P[E_{\tau_\alpha}] \leq 1$.

Since $\tau_\alpha := \inf\left\{n \geq 1 : E_n \geq \frac{1}{\alpha}\right\}$, we have $E_{\tau_\alpha} \geq \frac{1}{\alpha}$ and then

$$E_{\tau_\alpha} \geq \frac{1}{\alpha}\mathbf{1}_{\{\tau_\alpha < +\infty\}}.$$

Taking expectations gives

$$1 \geq \sup_{P\in\mathcal{P}} \mathbb{E}_P[E_{\tau_\alpha}] \geq \frac{1}{\alpha} \sup_{P\in\mathcal{P}} \mathbb{P}_P(\tau_\alpha \leq +\infty).$$

Therefore,

$$\sup_{P\in\mathcal{P}} \mathbb{P}_P\left(\tau_\alpha \le m\right) \le \alpha.$$

Then

$$\sup_{P\in\mathcal{P}} \mathbb{P}_P\left(\inf\left\{n\ge 1 : E_n \ge \frac{1}{\alpha}\right\} < +\infty\right) = \sup_{P\in\mathcal{P}} \mathbb{P}_P\left(\sup_{n\ge 1} E_n \ge \frac{1}{\alpha}\right) \le \alpha.$$

□

A single e-process starting at time 1 uses all observations from X_1 to X_n. If the changepoint occurs late, the value of this e-process may have become very small during the pre-change period. Consequently, after the change, the new observations must first compensate for the evidence accumulated before the changepoint.

To avoid this problem, we start a new e-process at each time j. The process starting at time j uses only the observations $X_j, \dots, X_n$ and is therefore not affected by observations before time j. Each starting time j represents a possible changepoint.

Definition 4 (e_j-process)**.** For any $j \ge 1$, let $\Lambda^{(j)} := \left\{\Lambda_n^{(j)}\right\}_{n\ge 1}$ be a nonnegative adapted process. The process $\Lambda^{(j)}$ is called an e_j-process for $\mathcal{P}$ if $\Lambda_1^{(j)} = \cdots = \Lambda_{j-1}^{(j)} = 1$, and

$$\sup_{P\in\mathcal{P}} \mathbb{E}_{P,\infty}\left[\Lambda_\tau^{(j)} \,\middle|\, \mathcal{F}_{j-1}\right] \le 1, \qquad \forall \tau \in \mathcal{T}.$$

The process $\Lambda^{(j)}$ starts accumulating evidence only from time j. When $j = 1$, $\Lambda^{(1)}$ is an ordinary e-process. Since the true changepoint is unknown, the collection $\Lambda^{(1)}, \Lambda^{(2)}, \dots$ is run in parallel and all these values are combined into one statistic used to trigger an alarm.

Definition 5. [E-detector] A nonnegative adapted process $M := \{M_n\}_{n\ge 0}$ is called an e-detector for the pre-change class $\mathcal{P}$ if

$$\mathbb{E}_{P,\infty}\left[M_\tau\right] \le \mathbb{E}_{P,\infty}\left[\tau\right], \qquad \forall \tau \in \mathcal{T}, \quad \forall P \in \mathcal{P}.$$

The difference between an e-process and an e-detector is important. For an e-process, the expected value at any stopping time is bounded by one. For an e-detector, the expected value is bounded by the expected stopping time (we will understant below that is because e-detector combines several e_j-processes, and each has expectation equal to 1)

Given an e-detector M, define

$$N^* := \inf\left\{n \ge 1 : M_n \ge \frac{1}{\alpha}\right\}.$$

Then the corresponding detection procedure satisfies

$$\inf_{P\in\mathcal{P}} \mathbb{E}_{P,\infty}\left[N^*\right] \ge \frac{1}{\alpha}.$$

Thus, thresholding an e-detector at $1/\alpha$ guarantees non-asymptotic ARL control.

Definition 6 (Shiryaev–Roberts and CUSUM e-detectors)**.** Let $\left\{\Lambda^{(j)}\right\}_{j\ge 1}$ be a sequence of e_j-processes. Set $M_0^{\mathrm{SR}} = M_0^{\mathrm{CU}} := 0$. For each $n \ge 1$, define the Shiryaev–Roberts e-detector and the CUSUM e-detector by

$$M_n^{\mathrm{SR}} := \sum_{j=1}^{n} \Lambda_n^{(j)}, \qquad M_n^{\mathrm{CU}} := \max_{1\le j\le n} \Lambda_n^{(j)}.$$

The Shiryaev–Roberts statistic sums the evidence from all possible starting times and the CUSUM statistic keeps only the strongest evidence among them.

Lemma 2. *Shiryaev–Roberts e-detector and the CUSUM e-detector introduced in Definition 6 are e-dectors.*

Proof. Let $\tau \in \mathcal{T}$ be a stopping time and let $P \in \mathcal{P}$ be a pre-change distribution.

If $\mathbb{P}_{P,\infty}(\tau = \infty) > 0$, then $\mathbb{E}_{P,\infty}[\tau] = \infty$, and the e-detector condition holds trivially. We may therefore assume that $\tau < \infty$ almost surely.

Since all e_j-processes are nonnegative, we have

$$M_\tau^{\mathrm{CU}} = \max_{1 \le j \le \tau} \Lambda_\tau^{(j)} \le \sum_{j=1}^{\tau} \Lambda_\tau^{(j)} = M_\tau^{\mathrm{SR}}.$$

It is therefore sufficient to prove the e-detector property for M^{SR}. Using nonnegativity and Tonelli's theorem,

$$\mathbb{E}_{P,\infty}\left[M_\tau^{\mathrm{SR}}\right] = \mathbb{E}_{P,\infty}\left[\sum_{j=1}^{\infty} \Lambda_\tau^{(j)} \mathbf{1}_{\{j \le \tau\}}\right] = \sum_{j=1}^{\infty} \mathbb{E}_{P,\infty}\left[\Lambda_\tau^{(j)} \mathbf{1}_{\{j \le \tau\}}\right].$$

Since τ is a stopping time, the event $\{j \le \tau\}$ belongs to $\mathcal{F}_{j-1}$. Hence,

$$\mathbb{E}_{P,\infty}\left[\Lambda_\tau^{(j)} \mathbf{1}_{\{j \le \tau\}}\right] = \mathbb{E}_{P,\infty}\left[\mathbf{1}_{\{j \le \tau\}} \mathbb{E}_{P,\infty}\left[\Lambda_\tau^{(j)} \,\middle|\, \mathcal{F}_{j-1}\right]\right] \le \mathbb{E}_{P,\infty}\left[\mathbf{1}_{\{j \le \tau\}}\right],$$

where the inequality follows from the defining property of an e_j-process.

Therefore,

$$\mathbb{E}_{P,\infty}\left[M_\tau^{\mathrm{SR}}\right] \le \sum_{j=1}^{\infty} \mathbb{E}_{P,\infty}\left[\mathbf{1}_{\{j \le \tau\}}\right] = \mathbb{E}_{P,\infty}\left[\sum_{j=1}^{\infty} \mathbf{1}_{\{j \le \tau\}}\right] = \mathbb{E}_{P,\infty}[\tau].$$

Combining this result with $M_\tau^{\mathrm{CU}} \le M_\tau^{\mathrm{SR}}$ gives

$$\mathbb{E}_{P,\infty}\left[M_\tau^{\mathrm{CU}}\right] \le \mathbb{E}_{P,\infty}\left[M_\tau^{\mathrm{SR}}\right] \le \mathbb{E}_{P,\infty}[\tau].$$

Thus, both M^{SR} and M^{CU} satisfy the e-detector condition. □

The associated stopping times correspond are

$$N_{\mathrm{SR}}^* := \inf\left\{n \ge 1 : M_n^{\mathrm{SR}} \ge c_{SR_\alpha}\right\}, \qquad N_{\mathrm{CU}}^* := \inf\left\{n \ge 1 : M_n^{\mathrm{CU}} \ge c_{CU_\alpha}\right\}.$$

The choice $c_\alpha = \frac{1}{\alpha}$ is always valid for controlling the ARL of the CUSUM e-detector. However, it may be conservative.

It remains to determine how the delayed-start processes $\Lambda^{(j)}$ can be constructed efficiently. For this purpose, we introduce a factor called a baseline increment.

Definition 7 (Baseline increment)**.** A nonnegative adapted process $L := \{L_n\}_{n \ge 1}$ is called a baseline increment for the pre-change class $\mathcal{P}$ if

$$\sup_{P \in \mathcal{P}} \mathbb{E}_{P,\infty}\left[L_n \,|\, \mathcal{F}_{n-1}\right] \le 1, \qquad \forall n \ge 1. \tag{6}$$

A baseline increment is not itself an e-process. It represents only the one-step multiplicative contribution used to construct a complete e_j-process.

Definition 8 (Baseline e_j-process)**.** Given a baseline increment L, define, for every $j \geq 1$,

$$\Lambda_n^{(j)} := \begin{cases} 1, & n < j, \\ \prod_{i=j}^{n} L_i, & n \geq j. \end{cases} \tag{7}$$

Under every pre-change distribution $P \in \mathcal{P}$, $\Lambda^{(j)}$ is a nonnegative supermartingale from time j onward. It is therefore a valid e_j-process.

Substituting this construction into the Shiryaev–Roberts and CUSUM e-detectors gives

$$M_n^{\mathrm{SR}} = \sum_{j=1}^{n} \prod_{i=j}^{n} L_i, \qquad M_n^{\mathrm{CU}} = \max_{1 \leq j \leq n} \prod_{i=j}^{n} L_i.$$

Definition 9 (Baseline SR and CUSUM e-detectors)**.** An SR or CUSUM e-detector constructed from baseline e_j-processes is called a baseline SR or baseline CUSUM e-detector.

These e-detectors admit the recursive representations

$$M_n^{\mathrm{SR}} = L_n \left(M_{n-1}^{\mathrm{SR}} + 1 \right), \tag{8}$$

$$M_n^{\mathrm{CU}} = L_n \max \left\{ M_{n-1}^{\mathrm{CU}}, 1 \right\}, \tag{9}$$

with $M_0^{\mathrm{SR}} = M_0^{\mathrm{CU}} = 0$.

Therefore, once a valid baseline increment L_n has been constructed, both statistics can be updated recursively without storing all previous e_j-processes.

The complete construction follows the logical order

$$L_n \quad \Longrightarrow \quad \Lambda_n^{(j)} = \prod_{i=j}^{n} L_i \quad \Longrightarrow \quad \begin{cases} M_n^{\mathrm{SR}} = \sum_{j=1}^{n} \Lambda_n^{(j)}, \\ M_n^{\mathrm{CU}} = \max_{1 \leq j \leq n} \Lambda_n^{(j)}, \end{cases} \quad \Longrightarrow \quad N^*.$$

2.2 Predictable plug-in construction for an unknown post-change distribution

In the online change-detection problem considered in the previous work, Algorithm 4 assumes that both the pre-change distribution P and the post-change distribution Q are known. However, in practice, the post-change distribution is generally unknown. Then our stategy is as following

We first choose a monitoring starting time t_0. The observations collected before t_0 are assumed to be pre-change observations and are used to estimate the pre-change parameter. Let $\widetilde{\theta}_0$ denote this estimate. Once monitoring begins, $\widetilde{\theta}_0$ is kept fixed.

At each monitoring time n, the unknown post-change parameter is estimated using only observations available before X_n is observed. The resulting estimator, denoted by $\widehat{\theta}_{1,n-1}$, is therefore $\mathcal{F}_{n-1}$-measurable. It can then be substituted into the one-step likelihood ratio.

Since $\widehat{\theta}_{1,n-1}$ is already known conditionally on $\mathcal{F}_{n-1}$, the resulting likelihood ratio still has conditional expectation equal to one under the fitted pre-change model. The following lemma formalizes this result.

Lemma 3 (Predictable plug-in baseline increment)**.** *Let $\widetilde{\theta}_0$ be the pre-change parameter estimated before the monitoring period and then treated as fixed. Assume that, under the fitted no-change model $P_{\widetilde{\theta}_0}$, the conditional distribution of X_n given $\mathcal{F}_{n-1}$ has density $p_{\widetilde{\theta}_0,n}(x \mid \mathcal{F}_{n-1})$.*

For every $\theta \in \Theta$, let $p_{\theta,n}(x \mid \mathcal{F}_{n-1})$ be a density. Let $\widehat{\theta}_{1,n-1}$ be an $\mathcal{F}_{n-1}$-measurable estimator of the post-change parameter. Define

$$L_n := \frac{p_{\widehat{\theta}_{1,n-1},n}(X_n \mid \mathcal{F}_{n-1})}{p_{\widetilde{\theta}_0,n}(X_n \mid \mathcal{F}_{n-1})}.$$

Then

$$\mathbb{E}_{\widetilde{\theta}_0,\infty}[L_n \,|\, \mathcal{F}_{n-1}] = 1$$

almost surely. Consequently, $\{L_n\}_{n\geq 1}$ is a valid baseline increment process for the fitted pre-change model.

Proof. Since $\widehat{\theta}_{1,n-1}$ is $\mathcal{F}_{n-1}$-measurable, it is fixed when conditioning on $\mathcal{F}_{n-1}$. Therefore,

$$\begin{aligned}\mathbb{E}_{\widetilde{\theta}_0,\infty}[L_n \,|\, \mathcal{F}_{n-1}] &= \int \frac{p_{\widehat{\theta}_{1,n-1},n}(x \mid \mathcal{F}_{n-1})}{p_{\widetilde{\theta}_0,n}(x \mid \mathcal{F}_{n-1})} p_{\widetilde{\theta}_0,n}(x \mid \mathcal{F}_{n-1})\ \mu(dx) \\ &= \int p_{\widehat{\theta}_{1,n-1},n}(x \mid \mathcal{F}_{n-1})\ \mu(dx) = 1.\end{aligned}$$

Hence, L_n satisfies the baseline-increment condition. □

The complete online procedure can now be described as follows.

1. Choose a monitoring starting time t_0. The change is assumed to occur after this time.
2. Use the observations collected before t_0 to estimate the pre-change parameter $\widetilde{\theta}_0$. This estimate is then kept fixed during monitoring.
3. Choose a window length W. At time n, estimate the post-change parameter using the most recent observations available before time n, for example,
$$\widehat{\theta}_{1,n-1} = \widehat{\theta}\left(X_{\max(1,n-W)}, \ldots, X_{n-1}\right).$$
4. Compute the baseline increment
$$L_n = \frac{p_{\widehat{\theta}_{1,n-1},n}(X_n \mid \mathcal{F}_{n-1})}{p_{\widetilde{\theta}_0,n}(X_n \mid \mathcal{F}_{n-1})}.$$
5. Construct the corresponding e_j-processes
$$\Lambda_n^{(j)} = \prod_{i=j}^{n} L_i, \qquad 1 \leq j \leq n.$$
6. Combine the e_j-processes into an e-detector, for example,
$$M_n^{\mathrm{SR}} = \sum_{j=1}^{n} \Lambda_n^{(j)} \text{ or } M_n^{\mathrm{CU}} = \max_{1\leq j\leq n} \Lambda_n^{(j)}.$$
7. Raise an alarm when the selected e-detector reaches its threshold.

In the next subsection, we apply this construction to the Poisson model considered in this work.

2.3 Example

2.3.1 Knowing λ_0 and λ_1

In the previous Poisson setting, where both the pre-change intensity λ_0 and the post-change intensity λ_1 were known (suppose that the parameter of the mark μ doesn't change. Let $X_n := N(t_n) - N(t_{n-1})$ denote the number of observed events during the interval $(t_{n-1}, t_n]$, and let $\Delta t_n := t_n - t_{n-1}$.

Under the pre-change model, $X_n \sim \text{Poisson}(\lambda_0 \Delta t_n)$, whereas under the post-change model, $X_n \sim \text{Poisson}(\lambda_1 \Delta t_n)$.

The baseline increment is defined as the one-step likelihood ratio

$$L_n := \frac{f_{\lambda_1 \Delta t_n}(X_n)}{f_{\lambda_0 \Delta t_n}(X_n)} = \exp\left\{-(\lambda_1 - \lambda_0)\Delta t_n\right\} \left(\frac{\lambda_1}{\lambda_0}\right)^{X_n},$$

where f_λ denotes the probability mass function of a Poisson distribution with mean λ. Indeed, under the pre-change model, the conditional expectation of the increment satisfies

$$\mathbb{E}_{\lambda_0}\left[L_n \,|\, \mathcal{F}_{n-1}\right] = \sum_{x=0}^{\infty} \frac{f_{\lambda_1 \Delta t_n}(x)}{f_{\lambda_0 \Delta t_n}(x)} f_{\lambda_0 \Delta t_n}(x) = \sum_{x=0}^{\infty} f_{\lambda_1 \Delta t_n}(x) = 1.$$

Therefore, L_n is a valid baseline increment.

2.3.2 Unknown post-change intensity

We now consider the case where the post-change intensity λ_1 is unknown. A monitoring starting time, denoted by n_0, is first selected. The observations collected before n_0 are assumed to come from the pre-change regime. They can therefore be used to estimate the pre-change intensity.

Let $\widehat{\lambda}_0 := \frac{\sum_{i=1}^{n_0} X_i}{\sum_{i=1}^{n_0} \Delta t_i}$ be the estimate of the pre-change intensity. Once monitoring starts, $\widehat{\lambda}_0$ is kept fixed.

At each monitoring time $n > n_0$, the post-change intensity is estimated using only past observations. Let W be a fixed window length and define $s_n := \max\{n_0 + 1, n - W\}$.

The post-change intensity is estimated by

$$\widehat{\lambda}_{1,n-1} := \frac{\sum_{i=s_n}^{n-1} X_i}{\sum_{i=s_n}^{n-1} \Delta t_i}.$$

Since $\widehat{\lambda}_{1,n-1}$ uses only observations available before time n, it is $\mathcal{F}_{n-1}$-measurable. The corresponding one-step likelihood-ratio increment is

$$L_n := \exp\left\{-\left(\widehat{\lambda}_{1,n-1} - \widehat{\lambda}_0\right)\Delta t_n\right\} \left(\frac{\widehat{\lambda}_{1,n-1}}{\widehat{\lambda}_0}\right)^{X_n}. \tag{10}$$

The estimator $\widehat{\lambda}_{1,n-1}$ is predictable because it does not use the current observation X_n. Therefore, conditionally on the past, it can be treated as fixed. By Lemma 3, L_n is a valid baseline increment with respect to the fitted pre-change model.

The corresponding e_j-process is

$$\Lambda_n^{(j)} = \prod_{i=j}^{n} L_i, \qquad n_0 + 1 \le j \le n.$$

These processes can then be combined into the Shiryaev–Roberts or CUSUM e-detector:

$$M_n^{\mathrm{SR}} = \sum_{j=n_0+1}^{n} \Lambda_n^{(j)}, \qquad M_n^{\mathrm{CU}} = \max_{n_0+1 \le j \le n} \Lambda_n^{(j)}.$$

An alarm is raised when the selected statistic reaches its threshold. For example,

$$N^* := \inf\left\{n > n_0 : M_n \ge c_\alpha\right\},$$

where the choice $c_\alpha = 1/\alpha$ guarantees the required ARL control when M_n is an e-detector.

Remark: For each candidate starting time j, a separate estimate of the post-change intensity can be computed. At time n, define $\widehat{\lambda}_{1,n-1}^{(j)} := \frac{\sum_{i=j}^{n-1} X_i}{\sum_{i=j}^{n-1} \Delta t_i}$.

This estimator uses only observations collected after the candidate starting time j and before the current observation X_n. Hence, each e_j-process has its own post-change intensity estimate.